

Daniel Gordon

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<https://danielgordon10.github.io/>

Education

The University of Washington	2014 – 2020
Ph. D. in Computer Science Advised by Dieter Fox and Ali Farhadi	
The University of Washington	2016
Masters in Computer Science	
Washington University in St. Louis	2010 – 2014
Bachelor of Science in Computer Science Second Major in Entrepreneurship Summa Cum Laude – GPA: 3.96, Engineering Class Rank: 8/323 Undergraduate Research Advised by Robert Pless	

Work Experience

Apple Maps 3DV:	Apr 2024 – Present
Senior Machine Learning Engineer - Responsible for vision language models, diffusion, semantic segmentation, and detection models in production at world scale - Developed end-to-end models for predicting 3D vectorized maps from raw pixels - Lead the data labeling pipeline effort of 10+ people across 4 multi-functional teams managing millions of samples, billions of labels, and hundreds of labelers	
Third Wave Automation:	
Engineering Lead: Perception - Oversaw a team of 5 engineers serving as both a manager and a contributor for several major projects - Facilitated cross-team collaboration resulting in doubling the success rate of autonomous workflows	Jun 2023 – Mar 2024
Tech Lead: Perception - Led a team of 3 engineers working on major features for the core competencies of the product - Contributed technical design expertise throughout the code stack - Pioneered Python type checking throughout the codebase	Jan 2022 – Dec 2022
Independent Contributor: Perception Designed and implemented many visual algorithms for the core product including object detection and tracking, pose estimation, local mapping and collision checking, and several safety systems	Jun 2020 – Dec 2021

Facebook AI Research (FAIR):

Winter 2019

Research Intern – A-STAR Team with Dhruv Batra

- Conducted research resulting in the ICCV publication “SplitNet: Sim2Sim and Task2Task Transfer for Embodied Visual Navigation”

Allen Institute for Artificial Intelligence:

Winter 2017

Research Intern – PRIOR Team with Roozbeh Mottaghi

- Conducted research resulting in the ICCV publication “Visual Semantic Planning using Deep Successor Representations”
- Collaborated and contributed to the development of AI2THOR

Google:

Software Engineering Intern – Google Maps

Summer 2013,
Summer 2014

- Designed and programmed the Street View Time Machine frontend
- Increased polish and feature improvement on the new Maps frontend

Engineering Practicum Intern – Google Wallet

Summer 2012

- Integrated an autocomplete feature to the Wallet website
- Added Google+ profile images and names to various Wallet pages
- Created the Wallet dashboard page and recent transaction widget

Publications**Learning by Watching and Learning by Doing**

2020

Ph.D. Thesis

Watching the World Go By: Representation Learning from Unlabeled Videos

Arxiv 2020

Daniel Gordon, Kiana Ehsani, Dieter Fox, Ali Farhadi

What Can You Learn from Your Muscles? Learning Visual Representations from Human Interactions

Arxiv 2020

Kiana Ehsani, Daniel Gordon, Thomas Nguyen, Roozbeh Mottaghi, Ali Farhadi

ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks

CVPR 2020

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, Dieter Fox

SplitNet: Sim2Sim and Task2Task Transfer for Embodied Visual Navigation

ICCV 2019

Daniel Gordon, Abhishek Kadian, Devi Parikh, Judy Hoffman, Dhruv Batra

What Should I Do Now? Marrying Reinforcement Learning and Symbolic Planning

Arxiv 2018

Daniel Gordon, Dieter Fox, Ali Farhadi

Shifting the Baseline: Single Modality Performance on Visual Navigation & QANAACL 2019
Short

Jesse Thomason, Daniel Gordon, Yonatan Bisk

IQA: Visual Question Answering in Interactive Environments

CVPR 2018

***Nvidia Pioneering Research Award**

Daniel Gordon, Aniruddha Kembhavi, Mohammad Rastegari, Joseph Redmon, Dieter Fox, Ali Farhadi

AI2-THOR: An Interactive 3D Environment for Visual AI Eric Kolve, Roozbeh Mottaghi, Daniel Gordon, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Matt Deitke, Kiana Ehsani, Yuke Zhu, Abhinav Gupta, Ali Farhadi	Technical Report 2017
Re3: Real-Time Recurrent Regression Networks for Object Tracking Daniel Gordon, Ali Farhadi, Dieter Fox	RA-L 2018
Visual Semantic Planning using Deep Successor Representations Daniel Gordon, Yuke Zhu, Eric Kolve, Dieter Fox, Li Fei-Fei, Abhinav Gupta, Roozbeh Mottaghi, Ali Farhadi	ICCV 2017
Collaborative Rephotography Ruth West, Abby Halley, Daniel Gordon, Jarlath O'Neil-Dunne, Robert Pless	SIGGRAPH 2013 Studio Talks
Collaborative Imaging of Urban Forest Dynamics: Augmenting Rephotography to Visualize Changes over Time Ruth West, Abby Halley, Jarlath O'Neil-Dunne, Daniel Gordon, Robert Pless	IS&T/SPIE 2013

Service

Co-organizer of 1st Workshop on Visual Understanding Across Modalities and THOR competition http://vuchallenge.org/	CVPR 2017
Organizer of Deep Learning in Practice Seminar Talk Series https://sites.google.com/cs.washington.edu/deeplearninginpractice/	Summer 2017

Teaching Experience

Teaching Assistant at the University of Washington Introduction to Deep Learning: Head TA - Wrote Numpy-based Pytorch-like library for deep learning assignments and autograder for grading - Managed 5 other TAs and 160 students	Fall 2018, Fall 2019
Teaching Assistant at Washington University in St. Louis Introduction to Artificial Intelligence Algorithms and Data Structures Logic and Discrete Mathematics Introduction to Computer Science	Spring 2013, Spring 2014 Fall 2013 Fall 2012 Fall 2010-Spring 2012

Honors and Awards

NVIDIA Graduate Fellowship (1 of 10 awardees from 230+ applicants)	2019
National Science Foundation GRFP Honorable Mention (Top 1/3 rd of applicants)	2015 and 2016

Wissner-Slivka Fellowship (University of Washington CSE)	2014
Achievement Rewards for College Scientists Fellowship (UW CSE 1 of 2 awardees)	2014-2016
Outstanding Senior Award – Computer Science (Washington University)	2014
Sigma Xi (Washington University)	Inducted Spring 2014
Upsilon Pi Epsilon (Washington University Top 1/3 rd of CSE Class)	Inducted Fall 2013
Tau Beta Pi (Washington University Top 1/8 th of Engineering Class)	Inducted Fall 2012

Patents

Methods and Apparatus for Checking The Size of Goods And Confirming Sufficient Room for Safe Placement of Goods in a Storage Location	March 9, 2025
Pending: Application US19/074,410	
Providing a thumbnail image that follows a main image	April 3, 2018
US Patent 9,934,222	
Display screen with graphical user interface or portion thereof	March 14, 2017
US Patent D780,795	

Technical Skills

Proficient in	Python, Java, PyTorch, C++, C, TensorFlow, Caffe, Git, HTML, CSS
Capable in	Bazel, Matlab, Javascript, Google Closure, Android, PHP, Mercurial, C#, LaTeX
Experience with	CUDA, Objective-C/Cocoa, iPhone, MySQL, JQuery, Unix Terminal

Open Source Repositories

VINCE: <https://github.com/danielgordon10/vince>
YouTUBE8M Data: <https://github.com/danielgordon10/youtube8m-data>
Deep Learning Class Numpy Library: <https://gitlab.com/danielgordon10/dl-class-2019a>
SplitNet: <https://github.com/facebookresearch/splitnet>
AI-Habitat: <https://github.com/facebookresearch/habitat-api>
AI2-THOR: <https://github.com/allenai/ai2thor>
Re3-Tensorflow: <https://github.com/danielgordon10/re3-tensorflow>
Re3-Pytorch: <https://github.com/danielgordon10/re3-pytorch>
IQA: <https://github.com/danielgordon10/thor-iqa-cvpr-2018>